

Justify your answers and show your work. Unless indicated otherwise, each problem is worth 10 points.

1. $\int \cos^3 2x \, dx$

$$\begin{aligned} &= \int \cos^2 2x \cos 2x \, dx \\ &= \int (1 - \sin^2 2x) \cos 2x \, dx \Big|_{u=\sin 2x, \, du=2 \cos 2x \, dx} \\ &= \int (1 - u^2) \frac{\cos 2x}{2 \cos 2x} \, dx \\ &= \frac{1}{2} \left(u - \frac{u^3}{3} \right) + C = \frac{1}{2} \sin 2x + \frac{\sin^3 2x}{6} + C \end{aligned}$$

2. $\int \sin^4 x \, dx$ (Hint: $\int \cos^2 2x \, dx = \frac{x}{2} + \frac{\sin(4x)}{8} + C$)

$$\begin{aligned} &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx \\ &= \frac{1}{4} \int 1 - 2 \cos 2x + \cos^2 2x \, dx \\ &= \frac{1}{4} \left(x - \sin 2x + \left[\frac{x}{2} + \frac{\sin(4x)}{8} \right] \right) + C \\ &= \frac{x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C \end{aligned}$$

3. $\int \sin^3 x \cos x \, dx$

$$\begin{aligned} &= \int \sin^2 x \sin x \cos x \, dx \\ &= \int (1 - \cos^2 x) \sin x \cos x \, dx \Big|_{u=\cos x \, du = -\sin x \, dx} \\ &= \int (1 - u^2) \frac{\sin x}{-\sin x} u \, dx = - \int (u - u^3) \, dx \\ &= - \left(\frac{u^2}{2} + \frac{u^4}{4} \right) + C = -\frac{\cos^2 x}{2} + \frac{\cos^4 x}{4} + C \\ &\quad \left(= \frac{\sin^4 x}{4} \right) \end{aligned}$$

4. $\int \frac{dx}{\sqrt{25 - x^2}}$ Let $x = 5 \sin \theta$ so $dx = 5 \cos \theta$,

$$\begin{aligned} &= \int \frac{5 \cos \theta}{5 \cos \theta} \, d\theta \\ &= \int 1 \, d\theta \\ &= \theta + C = \sin^{-1} \left(\frac{x}{5} \right) + C \end{aligned}$$

5. $\int_0^7 \sqrt{49 - x^2} \, dx$ Let $x = 7 \sin \theta$ so $dx = 7 \cos \theta$, $\sin^{-1} \left(\frac{7}{7} \right) = \frac{\pi}{2}$, and $\sin^{-1} \left(\frac{0}{7} \right) = 0$

$$\begin{aligned} &= \int_0^{\pi/2} 7 \cos x 7 \cos x \, dx \\ &= 49 \int_0^{\pi/2} \cos^2 x \, dx \\ &= 49 \left(\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right) \Big|_0^{\pi/2} = \frac{49\pi}{4} \end{aligned}$$

6. $\int \frac{x^2}{(\sqrt{25-x^2})^3} dx$ (Hint: $\int \frac{\sin^2 \theta}{\cos^2 \theta} d\theta = \int \tan^2 \theta d\theta = \tan \theta + \theta + C$) [5pts]

Let $x = 5 \sin \theta$ so $dx = 5 \cos \theta$,

$$\begin{aligned}
 &= \int \frac{25 \sin^2 \theta 5 \cos \theta}{(5 \cos \theta)^3} d\theta \\
 &= \int \frac{25 \sin^2 \theta}{25 \cos^2 \theta} d\theta \\
 &= \int \tan^2 \theta d\theta = \tan \theta + \theta + C \\
 &= \frac{x/5}{\sqrt{25-x^2}/5} + \sin^{-1} \left(\frac{x}{5} \right) + C = \frac{x}{\sqrt{25-x^2}} + \sin^{-1} \left(\frac{x}{5} \right) + C
 \end{aligned}$$

7. $\int \frac{x}{x^2+x-12} dx$ [15 pts]

$$\begin{aligned}
 \frac{x}{x^2+x-12} &= \frac{A}{x-3} + \frac{B}{x+4} \\
 x &= A(x+4) + B(x-3) \\
 \text{let } x=3 &\Rightarrow A = \frac{3}{7} \\
 \text{let } x=-4 &\Rightarrow B = \frac{4}{7}
 \end{aligned}$$

So,

$$\begin{aligned}
 \int \frac{x}{x^2+x-12} dx &= \int \frac{3}{7(x-3)} + \frac{4}{7(x+4)} dx \\
 &= \frac{3}{7} \ln |x-3| + \frac{4}{7} \ln |x+4| + C \\
 &= \ln^{1/7} |(x-3)^3(x+4)^4| + C
 \end{aligned}$$

8. $\int_0^{\ln 2} 4xe^x dx$ let $\begin{matrix} u=4x \\ du=4 dx \end{matrix}$ $\begin{matrix} dv=e^x dx \\ v=e^x \end{matrix}$ so,

$$\begin{aligned}
 &= 4xe^x \Big|_0^{\ln 2} - 4 \int_0^{\ln 2} e^x dx \\
 &= 4(xe^x - e^x) \Big|_0^{\ln 2} \\
 &= 4(\ln 2 \cdot 2 - 2) - 4(0 - 1) = 8 \ln 2 - 4
 \end{aligned}$$

9. $\int_1^\infty 6x^{-2} dx$

$$\begin{aligned}
 &= \lim_{b \rightarrow \infty} \int_1^b 6x^{-2} dx \\
 &= \lim_{b \rightarrow \infty} -\frac{6}{x} \Big|_1^b = 0 - (-6) = 6
 \end{aligned}$$

10. $\int_2^\infty \frac{17}{x \ln x} dx$

$$\begin{aligned}
 &= \lim_{b \rightarrow \infty} \int_1^b \frac{17}{x \ln x} dx \Big| \begin{matrix} u = \ln x \\ du = \frac{dx}{x} \end{matrix} \\
 &= \lim_{b \rightarrow \infty} \int_1^{\ln b} \frac{17}{x \cdot u} du = \lim_{b \rightarrow \infty} 17 \ln |\ln b| \Big|_1^b \\
 &= 17 \lim_{b \rightarrow \infty} (\ln |\ln b| - 0) = \infty
 \end{aligned}$$